

Student Development Task

Egypt Governorates Operations Dashboard

Overview

Build an interactive GIS web application using:

- ArcGIS Maps SDK for JavaScript v5
- ArcGIS Components
- Calcite Design System
- Vanilla JavaScript

The application will visualize **Egypt Governorates** and support **Operational issue reporting workflows**.

Business Scenario

The Ministry wants a lightweight operational GIS web application to:

- Visualize governorate **population** distribution
- Identify high-population regions
- Allow users to report:
 - Violating buildings
 - Street issues

The web application should provide both:

- **Visualization**
- **Operational reporting capabilities.**

Function Requirements

PART 1 — Governorates Visualization

Note: *This will require you to use the Governorate shapefile to create a new feature layer on ArcGIS Location Platform*

1. Display Egypt governorates FeatureLayer
2. Apply data-driven symbology based on **Population**
3. Provide meaningful popup information

Suggested Symbology

Use Class Break Renderer

(<https://developers.arcgis.com/javascript/latest/references/core/renderers/ClassBreaksRenderer/>)

Governorate Popup must display:

- Governorate Name
- Population
- Area (in KM)
- Population Density (optional)

PART 2 — Reporting System

Note: *This will require you to create a new feature layer on ArcGIS Location Platform*

The application should allow users to report:

1. Violating Buildings
2. Street Issues

Suggested Reporting Flow

Step 1: Select issue type

Step 2: Enter description

Step 3: Click map location

Step 4: Feature added to operational (Feature) layer

Feature Layer must have the following Fields

- **IssueType:** (Violating Building, Street Issue)
- **Description**
- **ReportedAt**

Different issue types must be displayed in different symbology (color and/or shape)

PART 3 — Filtering Reported Issues

The application should allow users to filter reported issues by type

User Flow:

- **Step 1: User** Select Issue Type from a dropdown list (this must be a separate dropdown list)
- **Step 2:** Reported issues are filtered on the map

Data Layers

Layer 1 — Governorates (based on the shared Shapefile)

Geometry: Polygon

Fields:

- Name_Ar
- Name_En
- Population
- Area_KM2 (to be added)

Layer 2 — Operational Issues (to be created)

Geometry: Point

Example Fields:

- IssueType
- Description
- ReportedAt

Stretch Goals (Bonus)

Add one or more of the following functions to the application:

- **Search widget (search for governorates)**
- **Legend Component**

- **LayerList Component**
- **Dashboard Counts:**
 - **Show counts of total reported issues**
 - **Show counts of reported issues within the visible extent**

The web application **MUST NOT** use any of the following:

- Editor Widget
- Frameworks (React/Angular/Vue)